

作品 3： 企業級多協定路 (Multi-Routing Protocols) 由重分發 (Redistribution) 與 GRE over IPsec 安全 VPN 整合實作

平台： Cisco IOS / GNS3

關鍵技術： Multi-Routing Protocols (EIGRP, OSPF, BGP), Route Redistribution, GRE over IPsec VPN

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作品商業背景概述

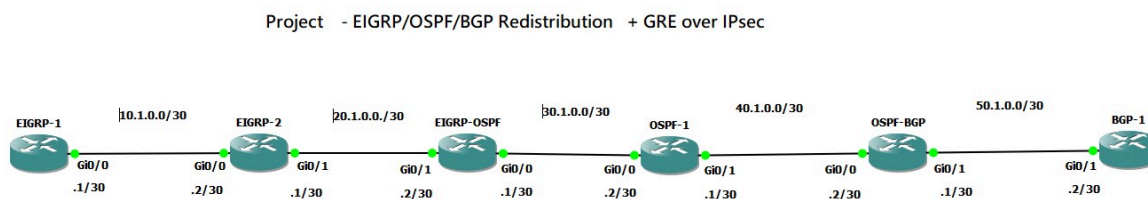
實務上，很少有公司的網路會只用一種路由協定。常見的原因像是：公司買下了別家企業，兩邊原本用的協定不同（比如 EIGRP 碰上 OSPF）；或者是對外連線到電信商時一定要跑 BGP；再加上有些舊設備只支援特定協定，這些因素都會讓企業網路變成多種協定混用的狀態。異構網路（多路由協定）難以互通，且跨公網傳輸缺乏安全加密與動態路由支援

系統架構

企業包含多個自治系統與區域 (EIGRP AS , OSPF Area , BGP AS) , 傳統點對點靜態設定擴充困難。本專案透過邊界路由器雙向路由重分發 (Route Redistribution) 實現全網動態互通, 並搭配 GRE over IPsec 架構封裝動態路由協定封包。

純 GRE 隧道缺乏加密能力 (明文傳輸) , 為了保護敏感企業資料, 疊加 **IPsec** (IKEv1/ESP) 進行機密性保護。同時關閉路由器與交換器上未使用的協定及遺留功能 (如關閉未使用的介面) 以增加安全性與防範網路攻擊

拓樸結構



IP位址表

EIGRP-1

Interface	IP address	Subnet Mask	Description
g0/0	10.1.0.1	/30	connect to router EIGRP-2
g0/1-7			shut down
Tunnel253	192.168.100.1	/30	tunnel to router BGP-1

EIGRP-2

Interface	IP address	Subnet Mask	Description
g0/0	10.1.0.2	/30	connect to router EIGRP-1
g0/1	20.1.0.1	/30	connect to router EIGRP-OSPF
g0/2-3			shut down

EIGRP-2

Interface	IP address	Subnet Mask	Description
g0/0	30.1.0.1	/30	connect to router OSPF-1
g0/1	20.1.0.2	/30	connect to router EIGRP-2
g0/2-3			shut down

OSPF-1

Interface	IP address	Subnet Mask	Description
g0/0	30.1.0.2	/30	connect to router EIGRP-OSPF
g0/1	40.1.0.1	/30	connect to router OSPF-BGP
g0/2-3			shut down

OSPF-BGP

Interface	IP address	Subnet Mask	Description
g0/0	40.1.0.2	/30	connect to router OSPF-1
g0/1	50.1.0.1	/30	connect to router BGP-1
g0/2-3			shut down

BGP-1

Interface	IP address	Subnet Mask	Description
g0/0			shut down
g0/1	50.1.0.2	/30	connect to router OSPF-BGP
g0/2			shut down
Tunnel253	192.168.100.2	/30	tunnel to router EIGRP-1

驗證 EIGRP Neighborhood

```
EIGRP-2#sh ip eigrp neighbors
```

EIGRP-IPv4 VR(EIGRP_1) Address-Family Neighbors for AS(100)									
H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RTO	Q Cnt	Seq Num	
1	20.1.0.2	Gi0/1	12	00:04:02	1	100	0	11	
0	10.1.0.1	Gi0/0	14	00:19:24	1	100	0	11	

驗證 OSPF Neighborhood

```
OSPF-1#sh ip ospf ne
```

```
OSPF-1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
50.1.0.1	1	FULL/BDR	00:00:33	40.1.0.2	GigabitEthernet0/1
30.1.0.1	1	FULL/DR	00:00:36	30.1.0.1	GigabitEthernet0/0

驗證 BGP Neighborhood

```
OSPF-1#sh ip ospf ne
```

```
OSPF-1#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
50.1.0.1	1	FULL/BDR	00:00:33	40.1.0.2	GigabitEthernet0/1
30.1.0.1	1	FULL/DR	00:00:36	30.1.0.1	GigabitEthernet0/0

驗證 package pass thought EIGRP, OSPF, and BGP

```
EIGRP-1#sh ip ro
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.1.0.0/30 is directly connected, GigabitEthernet0/0
L       10.1.0.1/32 is directly connected, GigabitEthernet0/0
D       20.0.0.0/16 is subnetted, 1 subnets
D       20.1.0.0 [90/15360] via 10.1.0.2, 02:06:55, GigabitEthernet0/0
D       30.0.0.0/30 is subnetted, 1 subnets
D EX    30.1.0.0 [170/1034240] via 10.1.0.2, 01:51:33, GigabitEthernet0/0
D       40.0.0.0/30 is subnetted, 1 subnets
D EX    40.1.0.0 [170/1034240] via 10.1.0.2, 01:50:27, GigabitEthernet0/0
D       50.0.0.0/30 is subnetted, 1 subnets
D EX    50.1.0.0 [170/1034240] via 10.1.0.2, 01:38:07, GigabitEthernet0/0
C       192.168.100.0/24 is variably subnetted, 2 subnets, 2 masks
L       192.168.100.0/30 is directly connected, Tunnel253
L       192.168.100.1/32 is directly connected, Tunnel253
EIGRP-1#
EIGRP-1#
EIGRP-1#trac
EIGRP-1#traceroute 50.1.0.2
Type escape sequence to abort.
Tracing the route to 50.1.0.2
VRF info: (vrf in name/id, vrf out name/id)
 0 10.1.0.2 3 msec 2 msec 1 msec
 1 20.1.0.2 4 msec 4 msec 4 msec
 2 30.1.0.2 4 msec 4 msec 4 msec
 3 40.1.0.2 5 msec 5 msec 4 msec
 4 50.1.0.2 6 msec 5 msec 5 msec
EIGRP-1#
```

驗證 packages send through tunnel

```
BGP-1#sh ip int br
Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       unassigned      YES NVRAM    administratively down  down
GigabitEthernet0/1       50.1.0.2        YES manual    up              up
GigabitEthernet0/2       unassigned      YES NVRAM    administratively down  down
GigabitEthernet0/3       unassigned      YES NVRAM    administratively down  down
NVI0                     50.1.0.2        YES unset    up              up
Tunnel253                192.168.100.2   YES manual    up              up
BGP-1#
BGP-1#
BGP-1#
BGP-1#ping 192.168.100.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.100.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/14 ms
```

驗證 IPsec Phase_1 (show crypto isakmp sa)

```
BGP-1#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst          src          state          conn-id status
10.1.0.1     50.1.0.2         QM_IDLE        1001 ACTIVE
```

驗證 IPsec Phase_2 (show crypto ipsec sa)

```
EIGRP-1#[A
% Bad IP address or host name
% Unknown command or computer name, or unable to find computer address
EIGRP-1#show crypto ipsec sa

interface: Tunnel253
  Crypto map tag: Tunnel253-head-0, local addr 10.1.0.1

  protected vrf: (none)
  local ident (addr/mask/prot/port): (10.1.0.1/255.255.255.255/47/0)
  remote ident (addr/mask/prot/port): (50.1.0.2/255.255.255.255/47/0)
  current_peer 50.1.0.2 port 500
    PERMIT, flags={origin is acl,}
    #pkts encaps: 10, #pkts encrypt: 10, #pkts digest: 10
    #pkts decaps: 10, #pkts decrypt: 10, #pkts verify: 10
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0
```